

August 7, 2026

Dear Hannah Handelman,

I am writing to apply for the Junior Software Engineer (Python) role you posted with Hunter Bond for your elite quant fund client. With a Master of Science in Computer Science from the University of Calgary and experience building Python/C++ software, backend APIs, data pipelines, and performance-sensitive research systems, I am excited by the opportunity to build tools that support real-time risk, pricing, and decision-making.

My background combines strong computer science fundamentals with practical software delivery. As a backend developer, I built Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL, implemented authentication and payment-service logic, and contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data. This experience gave me a solid foundation in Python services, relational data, user-facing systems, and maintaining reliability while translating product needs into working software.

I also bring research software experience from my M.Sc. work with BigGeo, Mitacs, and the University of Calgary. I developed Python and C++ workflows for large-scale data ingestion, preprocessing, storage, retrieval, and performance evaluation. My thesis work involved multiresolution data structures, wavelet methods, time-series approximation, and careful performance analysis, including improving encoding performance with C++ multithreading from 233 seconds to 26 seconds for a 10-tensor workload. That work strengthened the same analytical and mathematical problem-solving habits that I understand are important in front-office engineering.

The role interests me because it sits close to the people using the software: quants and traders relying on fast, scalable Python systems every day. I have not worked directly in financial markets yet, but I am comfortable learning complex technical domains quickly, asking precise questions, and turning new ideas into tested, documented tools. My teaching and research background also trained me to communicate clearly with technical stakeholders and to reason carefully about tradeoffs, correctness, and performance.

Thank you for considering my application. I would welcome the opportunity to discuss how my Python engineering, backend development, data-processing, and analytical systems experience could contribute to this front-office engineering team.

Sincerely,
Amir Mirzai Golpayegani
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